

Multi-period production and inventory

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Managerial motivation

Introduction: Managerial motivation

- ▶ A firm decides how much to **produce today** and how much to **keep in inventory** for future demand
- ▶ Producing in advance costs **holding**
- ▶ Producing at the last moment risks crashing into the **capacity** limit precisely in the peak months
- ▶ Problem class: **LP / convex QP**

The trade-off to balance

- ▶ The model balances four quantities:
 - ▶ margins
 - ▶ capacity
 - ▶ inventory
 - ▶ stability of the plan
- ▶ It is the ideal starting point of the laboratory: **every one of its numbers has an immediate economic interpretation**

Where this problem arises

- ▶ Every time a firm plans production over **several periods** with shared resources:
 - ▶ master production schedules (MPS)
 - ▶ aggregate planning
 - ▶ supply contracts with monthly profiles
- ▶ The core of the difficulty is the **intertemporal link**: today's decision (to produce in advance or not) changes tomorrow's constraints

In this chapter we will learn how to

1. formulate a **multi-period LP** with inventory balance constraints
2. read and check the **shadow prices** of capacity and their validity ranges
3. use a **quadratic term** to stabilize the plan
4. build the **capacity value curve**

Summary

- ▶ Decide how much to produce and how much to keep in inventory in every month
- ▶ The conflict is between the **holding cost** and the **capacity limit** in the peak months
- ▶ The intertemporal link prevents us from solving the months one at a time

Guided construction of the model

Introduction: Guided construction of the model

- ▶ We describe the problem in words, then we fix **data** and **variables**
- ▶ We write the minimum-cost production and inventory **LP model**
- ▶ We explain one by one the objective function and the families of constraints
- ▶ We will understand why the **inventory balance** is the heart of the model

The problem in words

- ▶ **We decide** how many units of each product to produce in each month and how many to keep in inventory
- ▶ **The objective** is to minimize the sum of the production and holding costs over the whole horizon
- ▶ **The constraints** are two:
 - ▶ the inventory accounting must add up in every month (what comes in equals what goes out, and all demand must be served)
 - ▶ the machine hours used in a month cannot exceed those available

Data (input of the model) (1/2)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of products; the products are indexed by $i \in \{1, 2, \dots, n\}$
m	$\in \mathbb{Z}_{\geq 1}$	number of months of the horizon; the months are indexed by $t \in \{1, 2, \dots, m\}$
d_{it}	$\in \mathbb{Q}_{\geq 0}$	demand for product i in month t (units)

Data (input of the model) (2/2)

Symbol	Type	Meaning
c_{it}	$\in \mathbb{Q}_{>0}$	production cost of one unit of i in month t (€)
h_{it}	$\in \mathbb{Q}_{>0}$	holding cost of one unit of i for month t (€)
a_i	$\in \mathbb{Q}_{>0}$	machine hours needed to produce one unit of product i
b_t	$\in \mathbb{Q}_{\geq 0}$	machine hours available in month t
s_{i0}	$\in \mathbb{Q}_{\geq 0}$	initial inventory of product i (units)

Decision variables

We introduce the following $2nm$ non-negative variables:

$$\begin{cases} x_{it} = \text{units of product } i \text{ produced in month } t \\ s_{it} = \text{units of product } i \text{ in stock at the end of month } t \end{cases}$$
$$\forall i \in \{1, 2, \dots, n\}, \forall t \in \{1, 2, \dots, m\}.$$

Using these variables, we write an LP model for the problem.

Minimum-cost production and inventory LP model

$$\min \sum_{i=1}^n \sum_{t=1}^m (c_{it} x_{it} + h_{it} s_{it}) \quad (4.1a)$$

$$\text{subject to } s_{i,t-1} + x_{it} = d_{it} + s_{it}, \quad \forall i \in \{1, 2, \dots, n\}, \forall t \in \{1, 2, \dots, m\}, \quad (4.1b)$$

$$\sum_{i=1}^n a_i x_{it} \leq b_t, \quad \forall t \in \{1, 2, \dots, m\}, \quad (4.1c)$$

$$x_{it} \geq 0, \quad \forall i \in \{1, 2, \dots, n\}, \forall t \in \{1, 2, \dots, m\}, \quad (4.1d)$$

$$s_{it} \geq 0, \quad \forall i \in \{1, 2, \dots, n\}, \forall t \in \{1, 2, \dots, m\}. \quad (4.1e)$$

Objective function and inventory balances

- ▶ The linear **objective function** minimizes the total cost of the plan: the sum of the production costs $c_{it} x_{it}$ and of the holding costs $h_{it} s_{it}$ over all products and months
- ▶ The **inventory balances**: for every product and every month, what comes in (previous inventory plus production) is equal to what goes out (demand served plus subsequent inventory)
 - ▶ for $t = 1$ the initial inventory s_{i0} is used
 - ▶ they are $n m$ linear constraints

Capacity and domain of the variables

- ▶ The **capacity constraints**

$$\sum_{i=1}^n a_i x_{it} \leq b_t$$

ensure that, in every month, the machine hours used by all the products do not exceed those available

- ▶ They are m linear constraints
- ▶ The constraints $x_{it} \geq 0$ and $s_{it} \geq 0$ define the variables of the model

Why the balance is the heart of the model

- ▶ It is the constraint that **links the months to one another**
- ▶ Without it the problem would break up into m **independent problems**
- ▶ And inventory would make no sense: nothing would connect what is left in the warehouse at the end of a month with the next one

What happens if... we remove the balance?

- ▶ The model **produces zero** (minimum cost!) because nothing forces it to serve demand
- ▶ If instead total demand exceeds the cumulated capacity, the model becomes **infeasible**; in that case either:
 - ▶ lost demand is allowed with a variable $u_{it} \geq 0$ and a penalty π_{it} in the objective
 - ▶ or capacity has to be revised

Summary

- ▶ Variables: x_{it} (production) and s_{it} (end-of-month inventory)
- ▶ Objective: **production** costs plus **holding** costs
- ▶ Constraints: **inventory balance** for every product and month, **capacity** in machine hours for every month
- ▶ The balance is what makes the problem genuinely **multi-period**

Worked example by hand

Introduction: Worked example by hand

- ▶ A minimal example: **one product** and **two months**
- ▶ We solve it entirely by hand, with no solver
- ▶ We also compute the **shadow price** of capacity by perturbation, and we will find the same mechanism again in the case study

Example: One product, two months (1/3)

Data:

- ▶ demand $d_1 = 60$, $d_2 = 100$
- ▶ capacity $b_1 = b_2 = 80$ units/month (with $a = 1$)
- ▶ production cost $c = 10$ €, holding $h = 1$ €/unit/month
- ▶ zero initial inventory

Step 1 — is it feasible? Total demand $160 \leq$ total capacity 160: yes, but only by producing at full capacity in *both* months.

Example: One product, two months (2/3)

Step 2 — balances.

$$x_1 = 60 + s_1, \quad s_1 + x_2 = 100$$

- ▶ With $x_2 \leq 80$ we need $s_1 \geq 20$, hence $x_1 \geq 80$
- ▶ But $x_1 \leq 80$: therefore $x_1 = 80$, $s_1 = 20$, $x_2 = 80$

Step 3 — cost.

$$10 \cdot (80 + 80) + 1 \cdot 20 = 1620 \text{ €}$$

The solution is **forced**: both capacity constraints are active.

Example: One product, two months (3/3)

Step 4 — shadow price by hand.

- ▶ With one more hour in month 2 ($b_2 = 81$): $x_2 = 81$, $s_1 = 19$, $x_1 = 79$
- ▶ The cost decreases by 1 € (one unit no longer sits in inventory for a month): the dual of the capacity of month 2 is -1 €/unit
- ▶ With one more hour in month 1, instead, nothing changes (month 1 already produces what is needed): dual 0

This mechanism — **the capacity of a month is worth as much as the holding it saves** — reappears identically in the case study.

Summary

- ▶ With tight capacity the solution can be **forced**: the balances determine it on their own
- ▶ The **shadow price** of a capacity is checked by adding one hour and recomputing the cost
- ▶ A **non-binding** capacity has zero shadow price
- ▶ The value of one more hour is the **holding it saves**

Case study

Introduction: Case study

- ▶ We apply the model to a realistic instance: **three products** over **six months**
- ▶ Demand is **seasonal and increasing**, with a peak in months 4–5, and the resource is single and shared
- ▶ We then see how the model is written in **Python with Gurobi**

The data of the products

Three products ($n = 3$) over six months ($m = 6$):

Product	c_i (€/u)	h_i (€/u/month)	a_i (hours/u)	s_{i0}
1	12.00	0.80	0.9	30
2	18.00	1.20	1.4	20
3	25.00	1.60	2.1	10

Product 3 is the most expensive one **and** the heaviest in machine hours: it is the one that will use the warehouse.

Capacity and monthly demand

month	1	2	3	4	5	6
b_t (hours)	420	420	460	460	460	420
$d_{1,t}$	110	130	150	190	210	160
$d_{2,t}$	70	80	110	140	150	100
$d_{3,t}$	50	55	60	80	95	70

- ▶ Data in the CSV files `produzione_domanda` and `produzione_capacita`
- ▶ Capacity grows only a little ($420 \rightarrow 460$ hours) while demand grows a lot

The model in Gurobi (1/2)

Variables and inventory balances:

```
m = gp.Model("produzione_scorte")
x = m.addVars(prodotti, mesi, name="x")    # produced
s = m.addVars(prodotti, mesi, name="s")    # in stock

# balance: incoming inventory + production
# = demand + outgoing inventory
m.addConstrs(
    ((s0[i] if t == 1 else s[i, t-1]) + x[i, t]
     == domanda[i, t] + s[i, t]
     for i in prodotti for t in mesi), name="bilancio")
```

The model in Gurobi (2/2)

Shared capacity and objective function:

```
# shared capacity (keep the constraints for the duals)
v_cap = m.addConstrs(
    (gp.quicksum(a[i] * x[i, t] for i in prodotti)
     <= capacita[t] for t in mesi), name="capacita")

m.setObjective(
    gp.quicksum(c[i]*x[i, t] + h[i]*s[i, t]
                for i in prodotti for t in mesi),
    GRB.MINIMIZE)
m.optimize()
```

Summary

- ▶ Instance: 3 products, 6 months, a single shared resource
- ▶ Seasonal demand with a peak in months 4–5, almost constant capacity
- ▶ In Gurobi: `addVars` for x and s , `addConstrs` for balances and capacity
- ▶ The capacity constraints are **kept in a variable** (`v_cap`) so that their duals can be read

Results

Introduction: Results

- ▶ We read the **optimal production plan** and the **inventory** month by month
- ▶ We look at the plots of the plan and of the inventory
- ▶ We will discover the strategy that the solver found all by itself: the **pre-build**

The optimal production plan

Optimal total cost: **32,889.33 €**

units/month	1	2	3	4	5	6
prod. 1	80.0	130.0	150.0	190.0	210.0	160.0
prod. 2	50.0	80.0	110.0	140.0	150.0	100.0
prod. 3	89.5	91.0	81.4	44.3	29.0	64.8

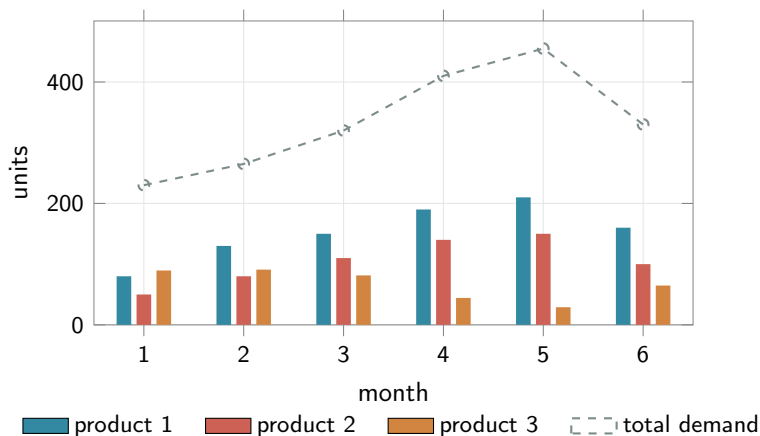
Products 1 and 2 follow demand; product 3 does not.

End-of-month inventory

units	1	2	3	4	5	6
prod. 1	0.0	0.0	0.0	0.0	0.0	0.0
prod. 2	0.0	0.0	0.0	0.0	0.0	0.0
prod. 3	49.5	85.5	106.9	71.2	5.2	0.0

- ▶ Only product 3 uses the warehouse
- ▶ The inventory goes to **zero** in the final month

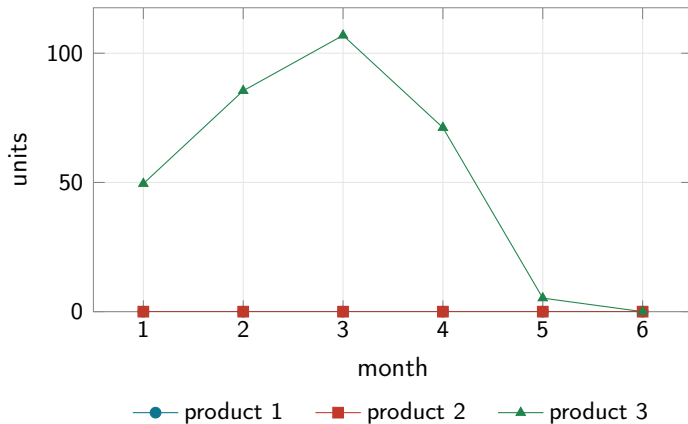
Optimal production plan and total demand



How to read the plot of the plan

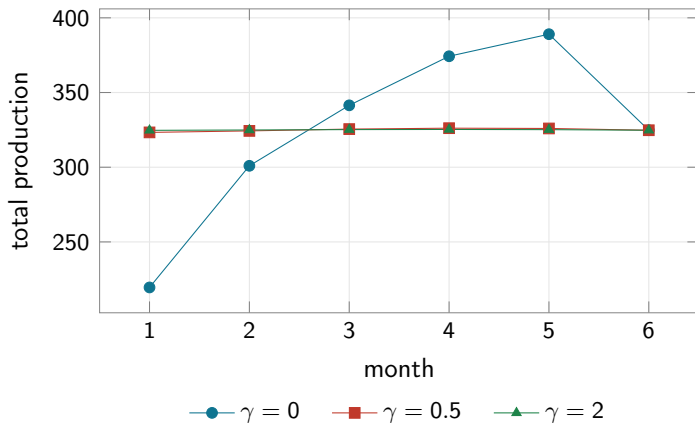
- ▶ The **bars** are the units produced of the three products in each month
- ▶ The **dashed line** is the total demand of the month
- ▶ In the peak months (4–5) total production stays **below** demand
- ▶ The difference is covered by the **inventory of product 3** accumulated in the first months

End-of-month inventory



Only product 3 uses the warehouse (**pre-build** strategy).

Effect of smoothing (γ)



Already with $\gamma = 0.5$ the production profile becomes almost **flat**.

The pre-build strategy (1/2)

- ▶ The solver discovered the *pre-build* strategy **all by itself**
- ▶ Products 1 and 2 (high holding costs relative to the machine hours they absorb) are produced **to demand**
- ▶ Product 3 — the most expensive one in machine hours — is produced in advance in months 1–3 and stockpiled (up to 107 units in inventory)
- ▶ In this way **capacity is freed up** in the peak months

The pre-build strategy (2/2)

- ▶ The inventory goes back to **zero** in the final month
- ▶ A positive terminal inventory would be wasted money, and the model knows it
- ▶ Nobody “taught” the pre-build to the solver: it is a consequence of the balances and of the capacity constraints

Summary

- ▶ Optimal total cost: **32,889.33 €**
- ▶ Products 1 and 2 produced to demand; product 3 in **pre-build**
- ▶ Inventory is a device to **move capacity** from the light months to the peak months
- ▶ Zero terminal inventory: emptying the warehouse at the end is always optimal

Sensitivity analysis

Introduction: Sensitivity analysis

- ▶ We read the **shadow prices** of capacity month by month, with their validity ranges
- ▶ We will derive an **exact** economic interpretation of them
- ▶ We finally build the **capacity value curve** and measure the cost of a regular plan

Shadow prices of capacity

month	use	Π (€/hour)	validity of b_t
1	330.0 / 420	0.000	$[330.0, +\infty)$
2	420.0 / 420	-0.762	$[330.0, 524.0]$
3	460.0 / 460	-1.524	$[370.0, 564.0]$
4	460.0 / 460	-2.286	$[370.0, 564.0]$
5	460.0 / 460	-3.048	$[399.0, 564.0]$
6	420.0 / 420	-3.810	$[330.0, 431.0]$

Check: +1 hour in month 6 takes the cost from 32889.33 to 32885.52 (delta -3.810).

How to read the shadow prices

- ▶ Month 1 is **not binding** (330 hours out of 420): its shadow price is zero
- ▶ All the other months are binding and have a **negative** shadow price: one more hour *reduces* the cost
- ▶ The value grows month after month: the most precious hour is the one of **month 6**
- ▶ The validity range says between which values of b_t the shadow price stays the same

Where the number 0.762 comes from (1/2)

- ▶ The duals grow by exactly 0.762 € per month: this is no accident

$$0.762 = \frac{h_3}{a_3} = \frac{1.6}{2.1}$$

- ▶ One more hour in month t allows us to produce $1/a_3$ units of product 3 *one month later*
- ▶ One month of holding is thus saved: $h_3 \cdot (1/a_3)$

Where the number 0.762 comes from (2/2)

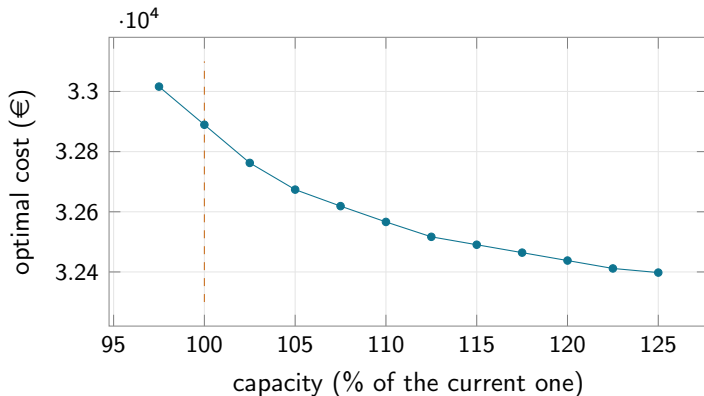
- ▶ The dual of month 6 is worth

$$5 \times 0.762 = 3.81 \text{ €/hour}$$

because one hour there avoids **five months** of holding for a unit produced in month 1

- ▶ When the solver and an elementary economic argument give the same number, the model is probably **right**

Capacity value curve



The dashed orange line marks the current capacity.

How to read the capacity value curve

- ▶ The curve is **convex and decreasing**: the first extra hours are worth a lot, then less and less
- ▶ Below **97%** of the current capacity the model becomes **infeasible**: demand can no longer be served in full
- ▶ This is managerial information as precious as the optimum itself

The cost of regularity: the QP variant

- ▶ The QP variant adds to the objective the **smoothing** term

$$\gamma \sum_{t=2}^m (X_t - X_{t-1})^2, \quad X_t = \sum_{i=1}^n x_{it}$$

- ▶ With $\gamma = 0.5$ the production profile becomes almost **flat**: the maximum variation drops from 81 to 1 units/month
- ▶ The additional cost is merely

$$33,003 - 32,889 = 114 \text{ €} \quad (+0.35\%)$$

The managerial recommendation

Typical recommendation from a lab report:

- ▶ It is worth buying **overtime hours** in months 5–6 up to 3–3.8 €/hour below the internal price
- ▶ A **levelled** plan costs only 0.35% more and drastically reduces the changes of pace
- ▶ Below **97%** of the current capacity full service cannot be guaranteed

Summary

- ▶ The shadow prices of capacity grow by $h_3/a_3 = 0.762$ € per month
- ▶ The non-binding month has zero shadow price; month 6 is the most precious one (3.81 €/hour)
- ▶ The capacity value curve is **convex and decreasing**, and it is interrupted at 97%
- ▶ The **regularity** of the plan costs very little: +0.35% with $\gamma = 0.5$

Exercises

Introduction: Exercises

- ▶ Five exercises to be carried out with the script of the case study
- ▶ They check the shadow prices, extend the model (lost demand, profit, emissions) and test a safety stock policy

Exercise: check

- ▶ Verify by perturbation the duals of months 3 and 4 (add one hour and reoptimize)
- ▶ Why is the dual of month 1 zero?

Exercise: lost demand

- ▶ Add the variable u_{it} (unserved demand) with penalty $\pi = 40$ €/unit
- ▶ Reduce capacity to 90%
- ▶ How much demand is lost, of which product, in which months? Interpret the choice of the model

Exercise: maximum profit

- ▶ Turn the model into a profit maximization with revenues $p = (20, 30, 42) \text{ €}$
- ▶ Make service optional with $y_{it} \leq d_{it}$
- ▶ Which demand is it worth *not* serving, and why?

Exercise: emissions

- ▶ Every unit of product 3 emits 2.5 kg of CO₂, products 1 and 2 one kg
- ▶ Add to the objective the term

$$\tau \sum_{i=1}^n \sum_{t=1}^m e_i x_{it}$$

- ▶ Draw the cost-emissions frontier for $\tau \in [0, 10]$ €/kg

Exercise: safety stock

- ▶ Impose the 10% safety stock:

$$s_{it} \geq 0.1 d_{i,t+1}$$

- ▶ How much does this policy cost?
- ▶ Which capacity constraint becomes the most critical?

Summary

- ▶ Check of the duals by perturbation
- ▶ Extensions of the model: **lost demand, maximum profit, emissions**
- ▶ Warehouse policies: **safety stock** and its cost

Final summary (1/2)

Model:

- ▶ Multi-period LP with production variables x_{it} and inventory variables s_{it}
- ▶ Inventory balance for every product and month, capacity in machine hours
- ▶ The balance is what links the months to one another

Optimal solution:

- ▶ Cost 32,889.33 €; pre-build of product 3 in months 1–3
- ▶ Zero terminal inventory

Final summary (2/2)

Economic reading:

- ▶ Shadow prices of capacity = holding saved, h_3/a_3 per month
- ▶ Capacity value curve convex and decreasing, infeasible below 97%

Stability of the plan:

- ▶ A quadratic smoothing term levels production
- ▶ With $\gamma = 0.5$ the plan is almost flat at a cost of +0.35%